



IMMUNOLOGIC (HIV) AND INFECTIOUS DISEASES

MODULE LEARNING OBJECTIVES

Describe the nursing management of clients with infectious diseases.

Use the nursing process and NCJMM as frameworks for care of the client with human immunodeficiency virus/acquired immunodeficiency syndrome.



**LIST AND DISCUSS VARIABLES THAT MAY
AFFECT THE FUNCTION OF A CLIENT'S
IMMUNE SYSTEM.**

ANSWER: VARIABLES THAT AFFECT IMMUNE FUNCTION (1 OF 2)

- **Age**

- Very young (immature system) and older adults – fewer and slower T cells, thinner skin, less stomach acid, weaker vaccine response.

- **Nutrition**

- Protein-calorie malnutrition and low zinc, iron, vitamins A, C, D and E reduce antibody and cell responses; obesity blunts them too.

- **Chronic disease**

- Diabetes, kidney or liver failure, cancer, autoimmune disease, HIV, burns, major trauma or surgery, splenectomy.

ANSWER: VARIABLES THAT AFFECT IMMUNE FUNCTION (2 OF 2)

- **Medications and treatments**

- Corticosteroids, chemotherapy, radiation, transplant immunosuppressants, biologics; long antibiotic courses disturb normal flora.

- **Lifestyle and environment**

- Chronic stress and poor sleep (cortisol), smoking, alcohol, inactivity, occupational and travel exposures, crowded living.

- **Other**

- Genetics and primary immunodeficiencies, pregnancy, and the immunization history that shapes acquired immunity.

INTERPRETATION OF LEUKOCYTE (WBC) VALUES

- What is the expected range for WBCs?
- Define leukopenia.
- Define leukocytosis.
- What is a “left shift” of WBCs?
- Define neutropenia.
- Contrast leukopenia and neutropenia.

ANSWER: INTERPRETING WBC VALUES (1 OF 2)

- **Expected WBC range**

- 5,000–10,000/mm³ (some texts 4,500–11,000). Neutrophils make up 55–70% of the total.

- **Leukopenia**

- Total WBC below 5,000/mm³ – too few white cells overall (marrow suppression, chemo, viral illness, some drugs).

- **Leukocytosis**

- Total WBC above 10,000/mm³ – usually infection or inflammation; also stress, steroids, leukemia.

ANSWER: INTERPRETING WBC VALUES (2 OF 2)

- **Left shift**

- More immature neutrophils (bands) pushed out of the marrow early – the sign of an acute bacterial infection (“bandemia”).

- **Neutropenia**

- Absolute neutrophil count (ANC) below 1,500/mm³; severe below 500. $ANC = WBC \times (\% \text{ segs} + \% \text{ bands})$.

- **Leukopenia vs neutropenia**

- Leukopenia = the whole count is low. Neutropenia = the neutrophils specifically are low, which is what drives infection risk. A client can be neutropenic with a near-normal total WBC and still needs protective precautions.

NEUTROPENIC PRECAUTIONS (PROTECTIVE ENVIRONMENT)



Neutropenic
Precautions

- What client population has the highest risk for developing neutropenia?
- Discuss the nurse's role when implementing neutropenic precautions.

ANSWER: NEUTROPENIA – WHO IS AT RISK AND THE NURSE'S ROLE

- **Highest risk**
 - Clients on chemotherapy (marrow suppression, nadir 7–14 days after a dose); also leukemia and lymphoma, bone-marrow or stem-cell transplant, radiation to marrow, advanced HIV, long-term immunosuppressants.
- **The nurse's role**
 - Private room; meticulous hand hygiene by everyone; screen visitors and staff – no colds, coughs or recent live vaccines.
 - No fresh flowers, plants or standing water; low-microbial diet per policy (no raw or unwashed produce, raw meat or eggs).
 - Protect the barriers: no rectal temps, suppositories or enemas; avoid IM injections and invasive procedures; soft-brush oral care; daily skin care; dedicated equipment kept in the room.
 - Monitor temperature every 4 hours and the ANC daily; signs are subtle because there are not enough neutrophils to make pus.
 - Fever of 100.4°F (38°C) is an emergency: cultures first, then IV broad-spectrum antibiotics within 1 hour of the order. Give filgrastim as ordered; mask the client when leaving the room.

MANAGEMENT OF CLIENTS WITH IMMUNE DEFICIENCY DISORDERS

HIV/AIDS

HIV is a chronic infection caused by the human immunodeficiency virus.

Acquired immunodeficiency syndrome (AIDS) is the disease caused by HIV infection.

HIV STATISTICS: US & WORLDWIDE

CDC - United States (2024 reports)

WHO HIV Worldwide Statistics

- About 1.2 million people are living with HIV; roughly 1 in 8 does not know it.
- An estimated 31,800 new infections in 2022 – down 12% from 2018, driven by a 30% drop among ages 13–24.
- 38,793 new HIV diagnoses in 2024; 69% of people with HIV had a suppressed viral load.
- **WHO / UNAIDS - worldwide (end of 2024)**

TRANSMISSION OF HIV

- Transmitted by body fluids containing infected cells
- Cannot be transmitted through casual contact.
- High-risk populations

HIV 101

Without treatment, HIV (human immunodeficiency virus) can make a person very sick and even cause death. Learning the basics about HIV can keep you healthy and prevent transmission.

HIV Can Be Transmitted By



Sexual Contact



Sharing Needles
to Inject Drugs



Mother to Baby
during pregnancy, birth,
or breastfeeding

HIV Is **NOT** Transmitted By



Air or Water



Saliva, Sweat, Tears, or
Closed-Mouth Kissing

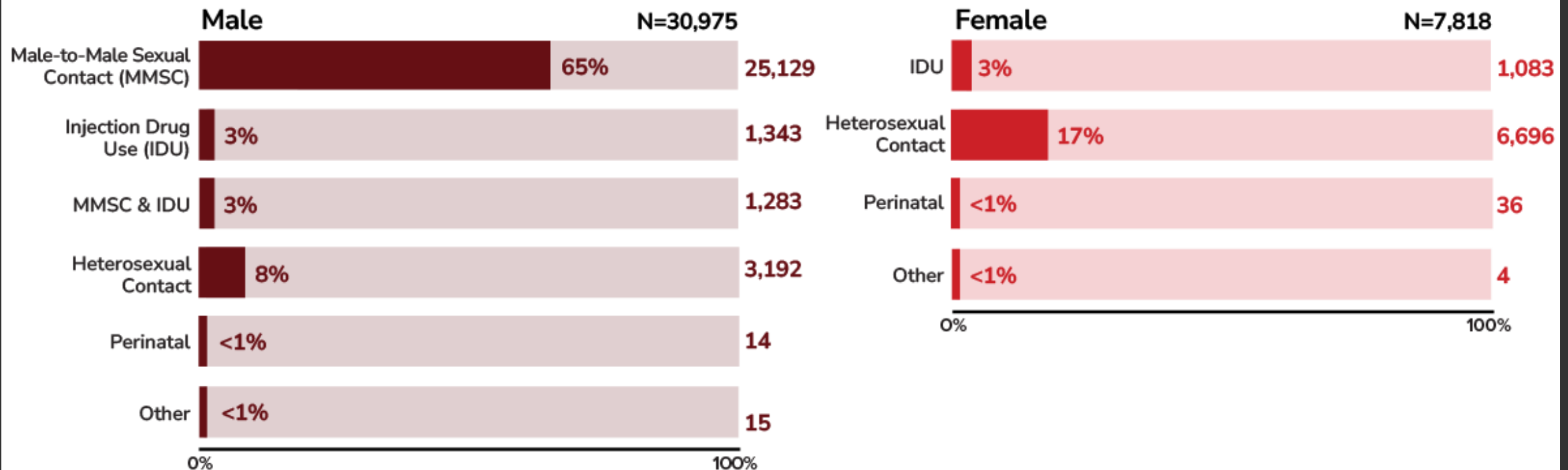


Insects or Pets



Sharing Toilets,
Food, or Drinks

HIV diagnoses by transmission category and sex, 2024–United States and 7 territories and freely associate states



Note. Data are presented for person aged ≥ 13 years at diagnosis.



OLDER ADULTS AND HIV

- In 2021, adults aged 50+ made up 16.3% of new HIV diagnoses in the US (CDC, 2023)
 - May not consider themselves to be at risk for HIV
 - Less likely than younger people to get tested
 - Normal age-related changes often mask the signs of HIV/AIDS
 - Clinical manifestations are like other illnesses that are common in older adults (infections such as pneumonia, cancers such as lymphoma)
 - More susceptible to HIV complications than younger adults
 - Older adult women have an increased risk due to the vaginal changes of aging
- The number of older adults living with HIV is increasing b/c

What education should the nurse provide about the prevention of HIV transmission?



Sex
without a condom



Sharing
injection equipment



Contaminated
blood transfusion



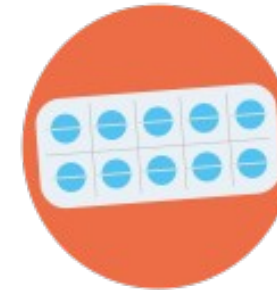
Passed
from mother to baby



Condom



U=U



PrEP

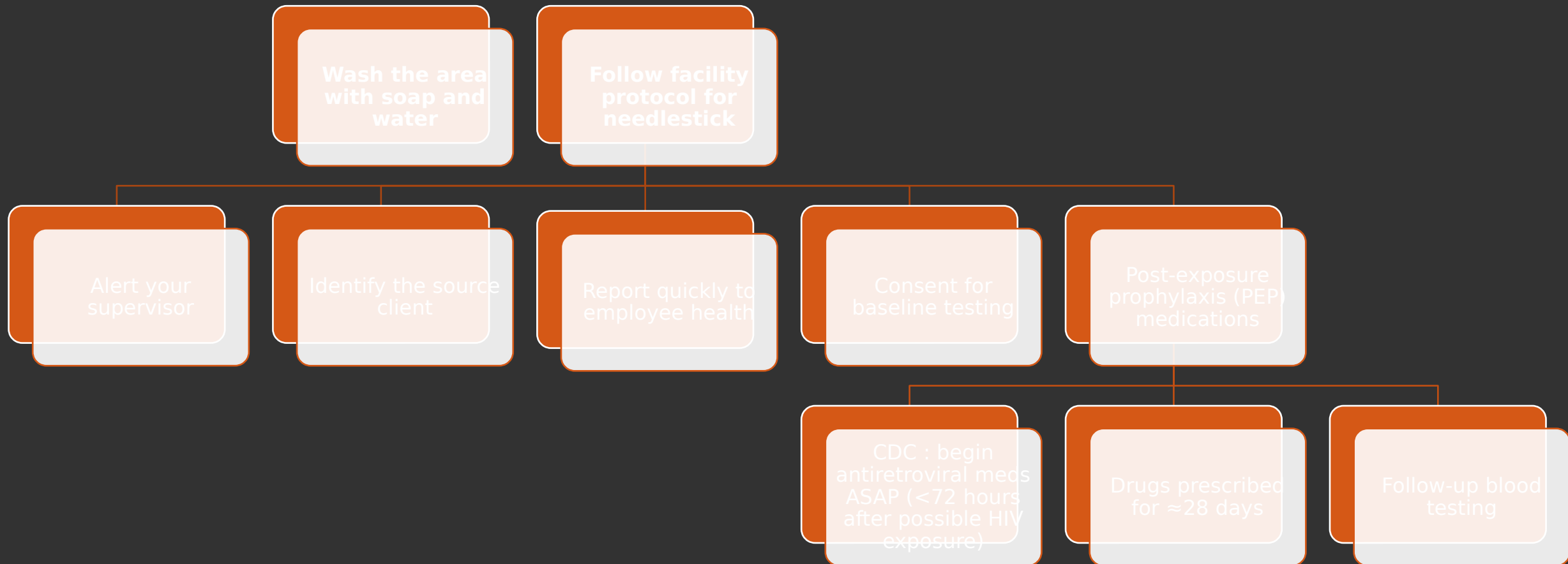


PEP

ANSWER: PREVENTING HIV TRANSMISSION – CLIENT EDUCATION

- Use a condom correctly every time (latex or polyurethane, water-based lubricant); limit partners; get tested with any new partner.
- Know your status: everyone 13–64 tested at least once, yearly or more if at risk. Treat other STIs – they make HIV easier to catch.
- PrEP for ongoing risk: daily oral emtricitabine/tenofovir or long-acting injectable cabotegravir, with HIV testing every 2–3 months.
- PEP after a possible exposure: start within 72 hours (sooner is better), three antiretrovirals for 28 days.
- U = U: a person on ART with an undetectable viral load does not transmit HIV sexually – adherence is prevention.
- Never share needles, syringes, works or razors; use syringe-services programs. Donated blood and organs are screened.
- Pregnancy: ART throughout, testing and prophylaxis for the newborn, formula rather than breastfeeding in the US – transmission falls below 1%.

OCCUPATIONAL EXPOSURE TO HIV (NEEDLE STICK)



STAGES OF HIV

Stage 0:
Primary
Infection like
a cold

Stage 1:
HIV
Asymptomatic
c for years

Stage 2:
HIV
Symptomatic
fevers,
weight loss,
swollen
nodes

Stage 3:
AIDS:
CD4<200, or
an AIDS
defining
disease

PRIMARY HIV INFECTION: STAGE 0

- From initial infection to the development of antibodies (stage 1)
- AKA acute HIV infection or window period
- Virus replicates rapidly in CD4 immune cells
- Usually present with non-specific viral illness symptoms that are similar to influenza
- Person can test negative for at least 2-3 weeks

HIV ASYMPTOMATIC: STAGE 1

- Client can remain asymptomatic for 10 years or more, but the immune system's CD4 cells are being destroyed
- Clients generally feel well and have few, if any, symptoms
- CD4 is within normal range (500 – 1500 cell/mm³)
 - High enough to preserve immune defensive responses

HIV SYMPTOMATIC: STAGE 2

- CDC Category B or stage 2
- HIV begins actively replicating
 - CD4+ cells are further destroyed
 - CD4 count = 200 – 499
 - Viral load increases
- Client develops symptoms/conditions related to the HIV infection

AIDS: STAGE 3

All people with AIDS have HIV, but not all people with HIV have AIDS.

- Defined as a CD4 count < 200 CD4 or the occurrence of an **AIDS defining illness**
- Not enough CD4 T-lymphocytes present to fight off infection
- Immune system is severely impaired
- Characterized by life-threatening opportunistic infections
- End stage of HIV
- Without treatment, death often occurs within 5 years



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graph LR; A[ELISA] --> B[Western blot]; B --> C[Viral load, CD4 count];
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ELISA

Western
blot

Viral load,
CD₄ count

HIV TESTING

HIV TESTING

- HIV antibody **screening**: Enzyme-linked immunosorbent assay (ELISA)
 - Many false positives, so NOT a definitive test
- HIV antibody **confirmation**: Western blot
- Viral load
 - Used for assessing effectiveness of antiviral medications
- CD4 count
 - Perform when Western blot is positive
 - Used for assessing the health of the client's immune system

HIV TESTING RESULTS

initial ELISA test is positive, it needs confirmation with Western blot t

- Positive results

- Psychological response
- Social and interpersonal consequences
- Begin treatment
- Ongoing counseling and education

- Negative results

- If ELISA test is negative, it may be repeated in weeks or months
- False sense of security
- Ongoing counseling and education

3 Things To Know *about* HIV Treatment



HIV medicines can't cure HIV, but they help people with HIV live longer, healthier lives.



People with HIV take a combination of HIV medicines every day. The HIV medicines prevent HIV from advancing to AIDS.



HIV medicines reduce the risk of HIV transmission. But people with HIV should still use condoms during sex.

For more information, visit: aidsinfo.nih.gov



TREATMENT FOR HIV INFECTION

Types of HIV Medications



Antiretroviral therapy

ART drugs are divided into 7 different classes

Combination HIV medicines

Often multiple types of ART medications are prescribed

PrEP (Pre-Exposure Prophylaxis)

For those who do not have HIV but are high risk

PEP (Post-Exposure Prophylaxis)

For those who may have been exposed to HIV in the past 72 hours

TREATMENT FOR HIV

- Treatment and protocols are continually evolving and have made tremendous advances in recent years
- Pharmacotherapy with antiretroviral therapy (ART)
- Goals of ART therapy
 - Suppress HIV replication
 - Reduce HIV-associated morbidity and prolong duration and quality of life
 - Restore and preserve immunologic function of client
 - Prevent HIV transmission by reducing serum viral load
 - Use of combination therapy to prevent resistant HIV infection

WHY IS ANTIRETROVIRAL THERAPY (ART) COMPLIANCE IMPORTANT?

If clients are non compliant with their ART, they may develop viral strains that are resistant to their current medication regimen.



If resistant strains develop, antiretroviral therapy may not be as effective in treating HIV/AIDS.

Risk of progressing from HIV+ to AIDS.

Increased risk of transmission.

COMBINATION DRUG THERAPY AS TREATMENT FOR HIV

- Multiple medication combinations are essential to prevent HIV resistance from developing
- Medications have been combined in capsules for easier administration and reduce non-adherence
- At least 2-3 medications are used in combination every day to suppress HIV replication



OVERVIEW OF MEDICATIONS USED IN ART

- Fusion inhibitors
 - Block the entry of HIV into cells
- Nucleoside reverse transcriptase inhibitors
 - Block the conversion of HIV RNA to DNA
 - Prototype: zidovudine (AZT)
 - Used in pregnant women to prevent passing the HIV to unborn child
 - Also given to newborns to protect from HIV mother
- Non-nucleoside reverse transcriptase inhibitors
 - Block the conversion of HIV RNA to DNA
- Protease inhibitors
 - Blocks proteases from forming the final HIV protein products

ART: NURSING MANAGEMENT

Monitor lab results

- CBC
- LFT
- CD4 count

Client education

- Adverse effects of medications
- Methods to decrease the severity of adverse effects
- Compliance
 - Take medications on a regular schedule and do not miss doses
 - Can lead to drug resistance

PULMONARY MANIFESTATIONS OF AIDS

- Pneumocystis (PCP) pneumonia
 - Most common AIDS-related opportunistic infection
 - Initial symptoms may be nonspecific
 - Progressive dyspnea, fever, nonproductive cough, chest discomfort
 - If untreated, progresses to respiratory failure
- Mycobacterium avium complex (MAC)
- Tuberculosis

What nursing assessments should be performed on the client's respiratory system?

ANSWER: RESPIRATORY ASSESSMENT IN AIDS

- Rate, depth, effort and pattern; accessory-muscle use; can the client speak in full sentences; position of comfort.
- SpO₂ at rest and with exertion – PCP classically desaturates with activity while the x-ray looks mild; ABGs when ordered.
- Auscultate all fields for crackles, diminished sounds and wheezes; note dyspnea on exertion and orthopnea.
- Cough: dry and nonproductive (PCP) versus productive; sputum color and blood; night sweats, fever pattern, weight loss (TB, MAC).
- Cyanosis of lips and nail beds; level of consciousness and restlessness as early hypoxia signs; chest discomfort.
- Screen for TB and keep airborne precautions until it is ruled out; collect sputum for PCP and AFB as ordered.
- Interventions: oxygen, semi-Fowler's, incentive spirometry, energy conservation, TMP-SMX (Bactrim) to treat PCP and to prevent it once CD4 is below 200.

GASTROINTESTINAL MANIFESTATIONS OF AIDS

- Candidiasis
 - Oral, esophageal, stomach
- Diarrhea from opportunistic infection
- Wasting syndrome
 - Involuntary loss of more than 10% of body weight and loss of muscle, plus 30 days of either diarrhea or weakness and fever

What nursing interventions and education should be provided for clients experiencing GI manifestations of AIDS?

ANSWER: GI MANIFESTATIONS OF AIDS – INTERVENTIONS AND EDUCATION

- Assess daily weight, intake and output, skin turgor and mucous membranes, stool number and character, albumin and electrolytes – potassium especially.
- Nutrition: small frequent high-calorie, high-protein meals, oral supplements, dietitian referral, appetite stimulants (megestrol, dronabinol) as ordered.
- Candidiasis: inspect the mouth daily; soft, cool, non-acidic foods; gentle oral care; nystatin swish-and-swallow or fluconazole.
- Diarrhea: fluids and electrolytes (IV if needed); low-fat, low-fiber, lactose-free diet; no caffeine or very hot or spicy food; antidiarrheals only after infection is ruled out; barrier cream for perianal skin.
- Food safety: no raw eggs, undercooked meat, unpasteurized dairy or juice, raw sprouts; wash produce; safe water; hand hygiene before eating.
- Wasting: weigh weekly, resistance exercise as tolerated, treat the underlying infection, control nausea before meals.

ONCOLOGIC MANIFESTATIONS OF AIDS



- Kaposi Sarcoma (KS)
 - Brownish pink to deep purple skin lesions
 - May be flat or raised
 - Surrounded by ecchymosis and edema
 - Lesions cause discomfort, disfigurement, ulceration, and potential for infection
- B-cell, Hodgkin, and non-Hodgkin lymphomas
 - Poor prognosis related the need for chemotherapy in a severely immunocompromised client

NEUROLOGIC MANIFESTATIONS OF AIDS

- HIV encephalopathy
 - Progressive cognitive, behavioral, and motor function decline
 - May lead to seizures
- Cryptococcus Neoformans
 - Fungal infection that causes fever, headache, malaise, stiff neck, N/V, mental status changes, seizures
- CMV (cytomegalovirus) retinitis
 - Herpes-type virus that can lead to visual impairment
- Depression: High prevalence in clients with HIV/AIDS

What nursing interventions and education should be provided for clients experiencing neurologic manifestations of AIDS?

ANSWER: NEUROLOGIC MANIFESTATIONS OF AIDS – INTERVENTIONS AND EDUCATION

- Assess level of consciousness, orientation, memory, speech, gait and balance, headache, neck stiffness, vision and seizure activity every shift.
- Safety: bed low and locked, call light in reach, help with walking, clear pathways, night light; seizure precautions with suction and oxygen at the bedside.
- Cognitive decline: one-step instructions, a consistent routine, calendar and clock, written medication schedules, pill organizers, family reminders.
- Cryptococcal meningitis: amphotericin B then fluconazole; dark quiet room; watch for rising intracranial pressure – worsening headache, vomiting, change in LOC.
- CMV retinitis: ganciclovir; report floaters or blurred vision at once; regular eye exams; adapt the home for low vision.
- Screen for depression and suicide risk at every visit; counseling, antidepressants, support groups; support caregivers; start advance-care planning early.

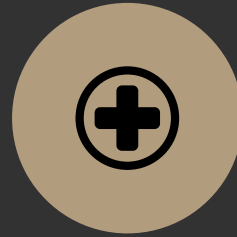
NURSING MANAGEMENT: OPPORTUNISTIC INFECTIONS



STRESS THE
IMPORTANCE
OF REGULAR
OFFICE VISITS



ASSESS FOR
MANIFESTATIONS OF
OPPORTUNISTIC
INFECTIONS,
INCLUDING
UNUSUAL
NEOPLASMS



IMMEDIATELY
REPORT
INDICATIONS
OF INFECTION
TO THE
PROVIDER



MONITOR
CD4 COUNT
TO ASSESS
THE LEVEL OF
REMAINING
IMMUNE
FUNCTION



MONITOR
VIRAL LOAD
LEVEL TO
ASSESS
EFFECTIVENESS OF THE
MEDICATION
REGIMEN

NURSING PLAN OF CARE

Develop a plan of nursing care for a client diagnosed with HIV infection. What are the priority nursing assessments and interventions? What education would you provide?

ANSWER: NURSING PLAN OF CARE FOR THE CLIENT WITH HIV

- **Priority assessments**

- Vital signs and temperature; breathing and SpO₂; weight, nutrition and hydration; skin, mouth and perianal area; neurologic status; CD4, viral load and CBC; medication list and adherence; mood, support system and safety at home.

- **Priority interventions**

- Protect from infection (hand hygiene, precautions, prophylaxis); give ART and other drugs on time; manage fever, pain, nausea, diarrhea and fatigue; keep up nutrition and skin integrity; balance activity and rest; refer to case management, social work and counseling.

- **Education**

- How HIV is and is not transmitted; ART adherence and side effects; signs of opportunistic infection to report; food and water safety; safer sex and U = U; PrEP for partners; vaccines; avoiding tobacco, alcohol and drugs; support groups and legal rights.



HIV/AIDS: EMOTIONAL & ETHICAL CONCERNS

HIPAA

Prejudices and
stigma

Fear of infection

Failure to disclose
HIV status

INFECTIOUS DISEASES

METHICILLIN-RESISTANT STAPHYLOCOCCUS AUREUS (MRSA)

- Easily colonizes on skin
- Individual serves as a reservoir for MRSA transmission to others
- Infection can spread to wounds or immunocompromised clients
- Treated with mupirocin ointment if not extensive, or IV vancomycin if severe or systemic
- Admission screenings
- Contact isolation recommended by CDC



VANCOMYCIN-RESISTANT ENTEROCOCCUS (VRE)

Normally found in the GI tract

Admission screenings done

Survives well on skin/hands and
environmental objects

2nd most common source of HAI in US

Contact isolation

CENTRAL LINE ASSOCIATED BLOOD STREAM INFECTION (CLABSI)



What nursing interventions should be implemented to prevent central line associated blood stream infections?

ANSWER: PREVENTING CLABSI – THE CENTRAL LINE BUNDLE

- Hand hygiene before every contact; maximal sterile barrier for insertion (cap, mask, sterile gown and gloves, full-body drape); chlorhexidine skin prep.
- Avoid the femoral site, use ultrasound guidance and an insertion checklist – the nurse is empowered to stop the procedure if sterility breaks.
- Scrub the hub with alcohol for 15 seconds and let it dry before every access; keep needleless connectors capped.
- Dressing: transparent semipermeable every 7 days, gauze every 2 days, either at once if wet, soiled or loose; chlorhexidine dressing; assess and date it every shift.
- Change IV tubing per policy (continuous every 96 hours, lipids every 24 hours, blood every 4 hours); daily chlorhexidine bathing in ICU.
- Ask every day whether the line is still needed and remove it as soon as it is not – line-days drive the infection rate.

CATHETER ASSOCIATED URINARY TRACT INFECTION (CAUTI)

**Table 1.
Four Essential Components
of CAUTI Prevention**

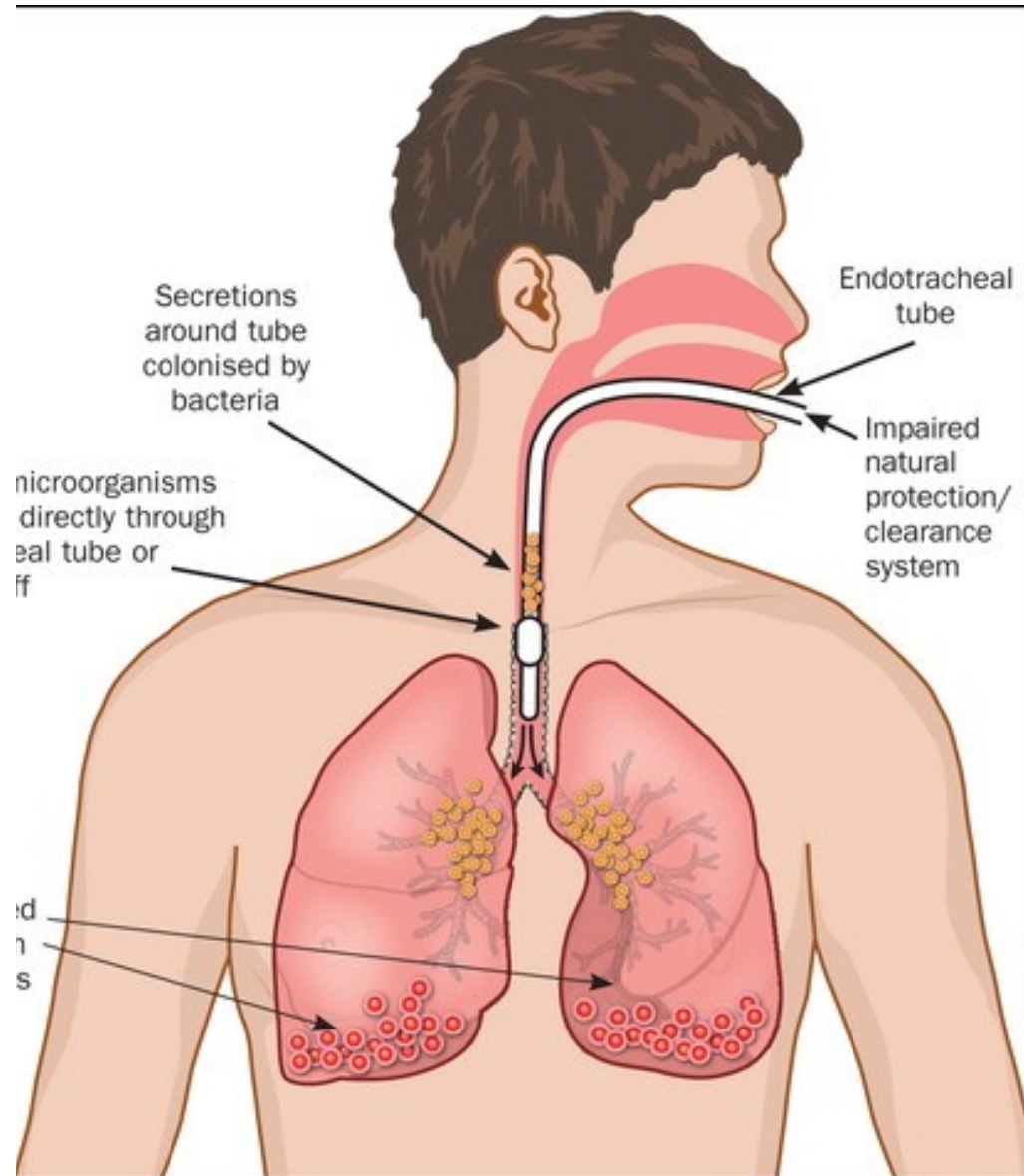
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|----|--|
| 1. | Avoid unnecessary placement of indwelling urinary catheters. |
| 2. | Use aseptic technique while inserting the catheter. |
| 3. | Use evidence-based guidelines for proper catheter management while in place. |
| 4. | Remove the catheter as soon as possible. |

What nursing interventions should be implemented to prevent catheter associated urinary tract infections?

ANSWER: PREVENTING CAUTI

- Insert only for a true indication – retention, exact output in the critically ill, protecting a perineal wound, comfort at end of life – never for incontinence or convenience.
- Aseptic insertion by trained staff, the smallest catheter that works, secured to the thigh with a securement device.
- Keep the closed system closed: bag below the bladder but off the floor, tubing free of kinks and dependent loops, empty into a clean container used only for that client, sample from the port.
- Daily perineal care with soap and water and after every bowel movement; hand hygiene and gloves for any handling.
- Assess need every day and remove early – nurse-driven removal protocols; use a bladder scanner, straight catheterization, or an external (condom or PureWick) catheter instead.
- Watch for fever, flank pain, cloudy or foul urine, and new confusion in older adults.

VENTILATOR ASSOCIATED PNEUMONIA (VAP)



What nursing interventions should be implemented to prevent ventilator associated pneumonia?

ANSWER: PREVENTING VAP – THE VENTILATOR BUNDLE

- Head of bed at 30–45 degrees at all times unless contraindicated.
- Oral care: brush teeth every 12 hours and give chlorhexidine or antiseptic oral care per policy every 2–4 hours; suction oral secretions.
- Daily sedation interruption and spontaneous breathing trial – extubate as soon as it is safe.
- Endotracheal tube with subglottic suctioning; cuff pressure 20–30 cm H₂O; drain circuit condensate away from the client; change circuits only when soiled.
- Hand hygiene and gloves; sterile technique for tracheal suctioning; turn every 2 hours; early mobility.
- DVT and stress-ulcer prophylaxis; avoid gastric overdistension with tube feeds; check tube placement and residuals.

Significant cause of mortality worldwide, especially in children (CDC, 2020j)

Priority concern: Dehydration

Most effective treatment:
Hydration

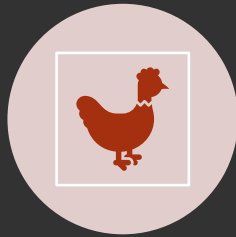
Primarily transmitted through oral ingestion

DIARRHEAL DISEASES

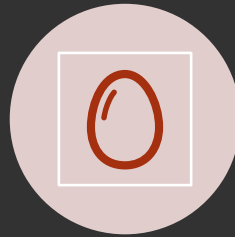
COMMON DIARRHEAL DISEASES



CALICIVIRUS
(NOROVIRUS)



CAMPYLO-
BACTER



SALMONELL
A



ESCHERICHI
A COLI



GIARDIA
LAMBLIA

DIARRRHEAL DISEASES: NURSING MANAGEMENT

What nursing assessments should be done
and what interventions should be
implemented to treat and prevent diarrheal
disease?

ANSWER: DIARRHEAL DISEASE – ASSESSMENT, TREATMENT AND PREVENTION

- **Assess**

- Stool frequency, volume, blood or mucus; onset, food and travel history, recent antibiotics; vital signs and orthostatics, weight, skin turgor, mucous membranes, urine output; potassium, sodium, BUN and creatinine; stool culture, ova and parasites, C. difficile toxin.

- **Treat**

- Oral rehydration solution, IV fluids for moderate to severe loss; replace potassium; contact (enteric) precautions with soap-and-water hand washing; skin care; bland diet as tolerated, no caffeine, alcohol, dairy or high-fat foods; antimotility drugs only when there is no fever, blood or toxin; antibiotics only for specific organisms.

- **Prevent**

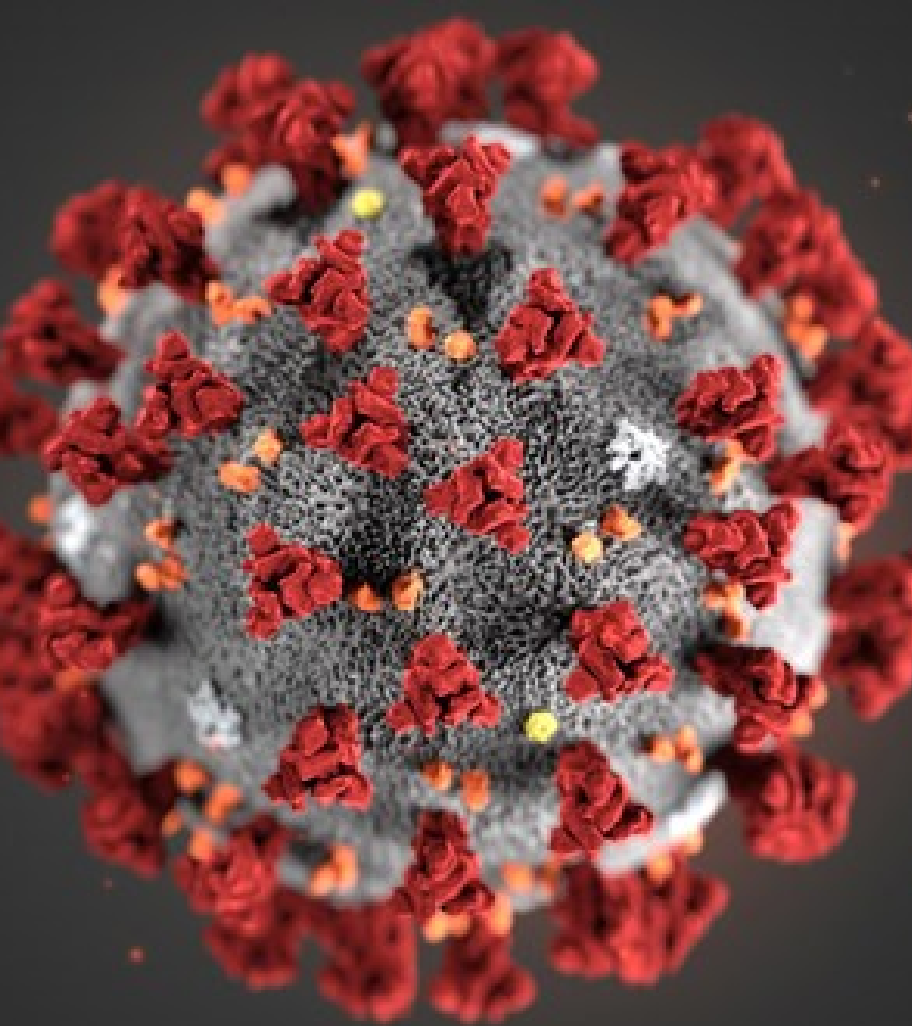
- Hand hygiene after toileting and before food; cook meat fully, wash produce, refrigerate leftovers, safe water when traveling; rotavirus vaccine for infants; antibiotic stewardship and limited PPIs to prevent C. difficile; report outbreaks to the health department.

EMERGING INFECTIOUS DISEASES

- Zika virus – global reduction in worldwide cases since 2017
- West Nile virus – leading cause of mosquito-borne disease in the U.S.
- Ebola virus disease – rare but severe, often fatal illness (50%)
 - Worldwide outbreak in 2014
- Legionnaires Disease – multisystem illness that includes pneumonia and respiratory symptoms



EMERGING INFECTIOUS DISEASES: COVID-19



- SARS-CoV-2 is primarily transmitted by close contact through droplets and aerosolized particles.
- CDC recommendations include all PPE measures be taken including contact, airborne, and droplet precautions.
- Primary prevention methods, like vaccination, should be included in nursing education.
- Adults aged 65+ years, pregnancy, immunosuppression, and chronic lung disease increase risk of severe disease
- Wide spectrum of clinical manifestations
- About 80% of clients with mild symptoms can be managed at home
- Severe disease may require oxygen and ventilatory support

IMMUNIZATIONS

IMMUNIZATIONS

- Administration of vaccine causes production of specific protective antibodies
- Contraindications
 - Previous serious reaction to vaccine
 - Some live vaccines contraindicated in severely immunosuppressed individuals and during pregnancy
- Adult vaccines
 - **Influenza - yearly**
 - Herpes zoster (shingles)
 - COVID-19
 - Tetanus
 - Pneumococcal
 - Meningococcal
 - Hepatitis A & B
 - All healthcare workers should have documented immunity from MMR

[CDC Adult Immunization Schedule by Age](#)



QUESTIONS?

FOR REVIEW:
ISOLATION PRECAUTIONS

RESOURCES FOR INFECTION CONTROL

- World Health Organization (WHO)
- Centers for Disease Control and Prevention (CDC)
- Occupational Safety and Health Administration (OSHA)
- The Joint Commission (TJC)
- Local agencies (health departments)
- Hospital/facility infection control specialists and facility policies



Guidelines to prevent the transmission of microorganisms in hospitals

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graph TD; A[Guidelines to prevent the transmission of microorganisms in hospitals] --> B[Standard precautions used for all clients]; B --> C[Transmission-based precautions are for clients with known infectious diseases spread by airborne, droplet, or contact routes]; C --> D[ISOLATION PRECAUTIONS]
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Standard precautions used for all clients

- *The primary strategy for preventing HAIs*

Transmission-based precautions are for clients with known infectious diseases spread by airborne, droplet, or contact routes

ISOLATION PRECAUTIONS

STANDARD PRECAUTIONS

Hand hygiene

Use of gloves and other barriers (PPE)

Proper handling of client care equipment and linen

Environmental controls

Prevention of injury from sharps and needles

Clients' room assignments within facility

These are used for ALL clients!



AIRBORNE CONTACT PRECAUTIONS

Visitors check in with Nursing before entering.

RESTRICTED VISITATION

ALL PPE REQUIRED REGARDLESS OF VACCINATION STATUS OR NATURAL IMMUNITY

PRIOR TO ENTERING:



Wash or gel hands



Wear gown and gloves



Use a NIOSH respirator
(N95/PAPR/CAPR)

OTHER REQUIREMENTS:



High priority for Airborne-Infection Isolation Room if available, keep door closed.



Wear eye protection, face shield or goggles if contact with secretions are likely.



Use patient-dedicated or disposable equipment.
Clean and disinfect shared equipment.

Sign to be removed by Environmental Services after precaution discontinuation and room cleaned.

AIRBORNE PRECAUTIONS

- For clients with airborne pathogens
 - TB
 - Varicella
 - Measles
- Private, negative air pressure/airflow room
- Keep door closed
 - Negates room pressure if left open
- N-95 mask

DROPLET PRECAUTIONS



DROPLET PRECAUTIONS

Visitors check in with Nursing before entering.

For source control, ask patient to don mask if tolerated when healthcare workers/visitors are present.

PRIOR TO ENTERING:



Wash or gel hands



Wear mask



Wear eye protection
Face shield or goggles

OTHER REQUIREMENTS:



Must use patient dedicated or disposable equipment.
Clean and disinfect shared equipment.

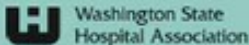


If performing an Aerosol Generating Procedure (AGP), consider wearing a higher-level protection mask or respirator and keep door closed. Follow Facility Policy for performing AGPs.



Wear gown and gloves if contact with secretions is likely.

Sign to be removed by Environmental Services after precaution discontinuation and room cleaned.



Washington State
Hospital Association

4/10/2023



Washington State Department of
HEALTH

- Transmission by close contact
- Influenza, meningitis, mumps, pertussis
- Face mask used within 3 to 6 feet of the client
- Private room
- Door can remain open



CONTACT PRECAUTIONS

Visitors check in with Nursing before entering.

PRIOR TO ENTERING:



Wash or gel hands



Wear Gown and gloves

OTHER REQUIREMENTS:



Use patient dedicated or disposable equipment.
Clean and disinfect shared equipment.

Sign to be removed by Environmental Services after precaution discontinuation and room cleaned.

CONTACT PRECAUTIONS

- Spread by skin-to-skin contact
 - MRSA, VRE
- Private room, or room with client who has same infection
- Gowns
- Masks not needed
- Doors do not need to be closed
- Organism is easily transmitted from client to healthcare worker

CLOSTRIDIUM DIFFICILE (C. DIFF)



CONTACT ENTERIC PRECAUTIONS

Visitors check in with Nursing before entering.

PRIOR TO ENTERING:



- Wash or gel hands prior to entry.
- Use SOAP AND WATER upon leaving the room.



Wear a gown and gloves.

OTHER REQUIREMENTS:



Use patient-dedicated or disposable equipment.

Clean and disinfect shared equipment before leaving the room with sporicidal/bleach cleaner.

Sign to be removed by Environmental Services after precaution discontinuation and room cleaned.

- Related to antibiotic therapy or immunosuppression
- Toxins from bacteria are released into the bowel and cause excessive diarrhea
- Spreads easily
- Treated with oral vancomycin or metronidazole
- Special contact precautions
- Use gowns and gloves for all client contact
- Handwashing with soap and water
- Clean equipment with a bleach product